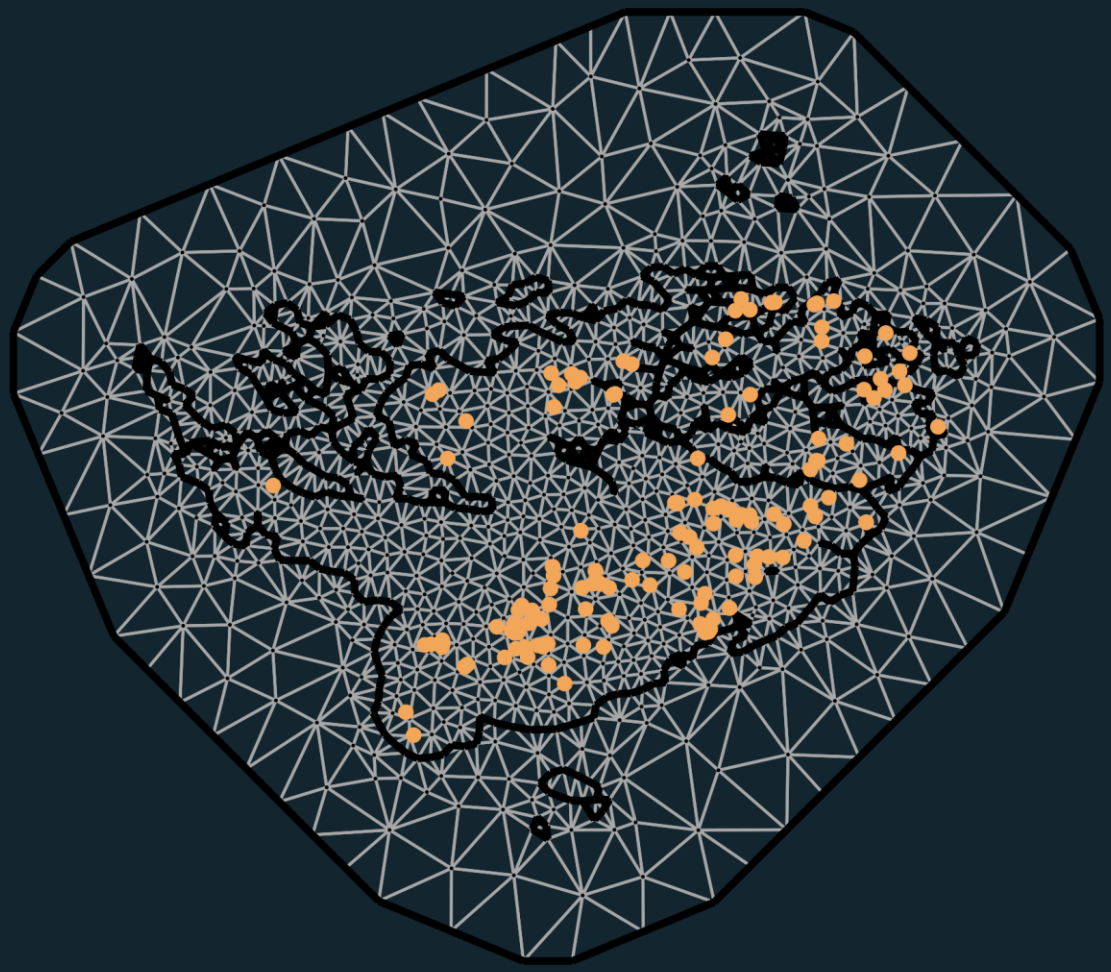
Bayesian Spatial Prediction with R-INLA

Application to sulfate mapping in groundwater

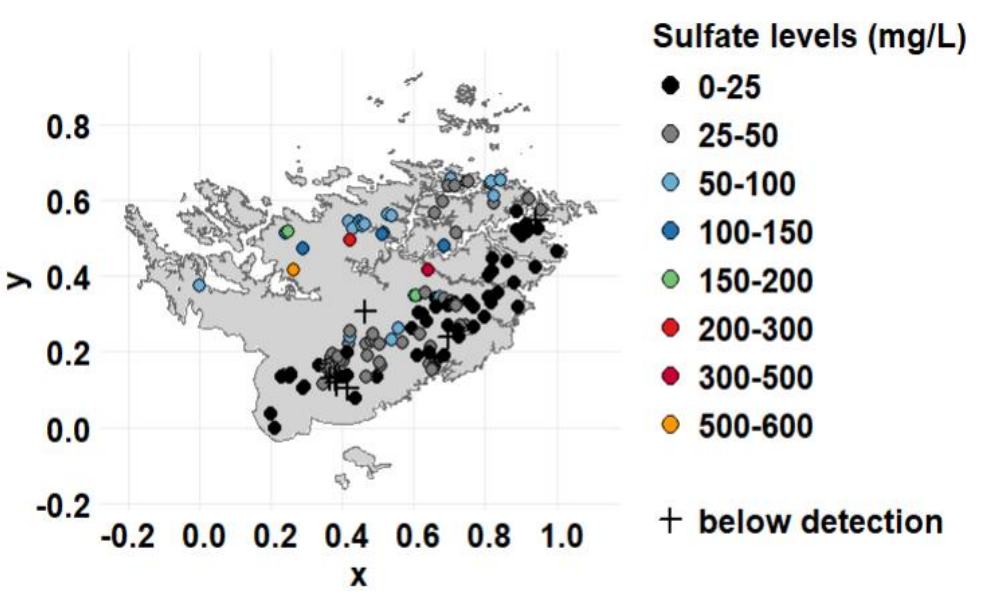


Dr. Corinna Perchtold

Bureau de Recherches Géologiques et Minières (BRGM) supported by the French National Research Agency within the HOUSES project (grant N°ANR-22-CE56-0006), Orléans

THE PROBLEM

We see points. We want a surface.



148 groundwater sulfate measurements in a domain of interest. What is the concentration in-between when accounting for covariates?

Setup:

- Bayesian framework

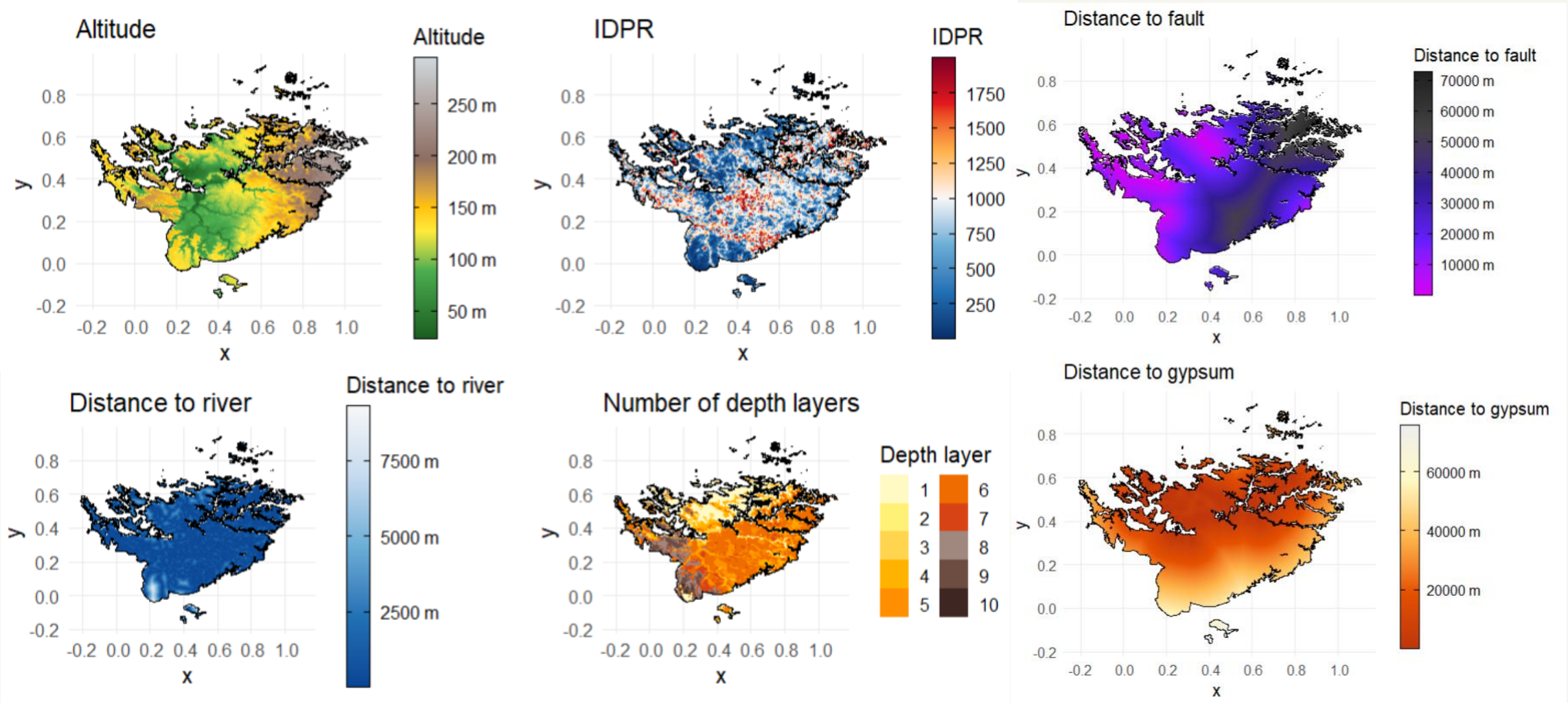
Goal:

- Spatial prediction map with posterior covariate estimates AND
- Uncertainty for every grid cell

Open plot_me.R to plot the data.

THE FIXED EFFECTS

The covariates



THE MODEL

A geostatistical model

$$y(s_i) = \beta_0 + \beta_1 \cdot altitude(s_i) + \beta_2 \cdot depth\ layer(s_i) + \beta_3 \cdot IDPR(s_i) + \beta_4 \cdot dist\ to\ river(s_i) + \beta_5 \cdot dist\ to\ fault(s_i) + \beta_6 \cdot dist\ to\ gypsum(s_i) + u(s_i) + \varepsilon(s_i)$$

Fixed effects $\boldsymbol{\beta}$: Intercept β_0 and covariates β_1 to β_6 .
Here: altitude, depth layer, IDPR, distance to the river, distance to fault and distance to gypsum.

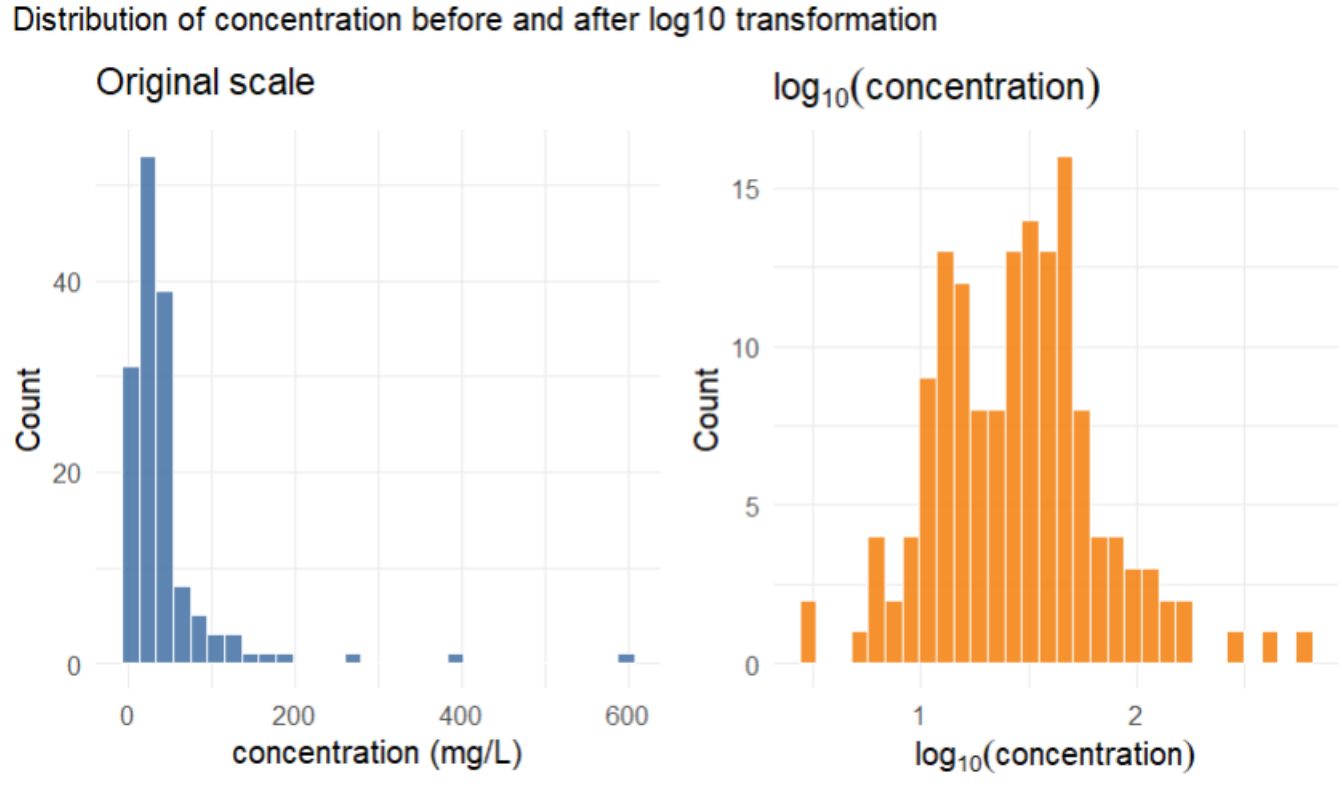
Spatial field $\boldsymbol{u(s)}$: captures spatially correlated variation that $\boldsymbol{\beta}$ do not explain. Nearby locations take similar values, and the similarity fades with distance. Range ρ sets how far it correlates, and standard deviation σ how strong it is.

Noise $\boldsymbol{\varepsilon(s)}$: Measurement error, $N(0, 1/\tau)$. Precision τ.

We place the geostatistical model in a Bayesian framework.
The same framework covers GLMMs, GAMs, time series and survival models.

THE DATA PROPERTIES

Check distribution of data



Open model_setup.R file and start with Step 0.

Why log10?

Groundwater sulfate is left-skewed.

On the log10-scale a Gaussian likelihood fits well.

WHAT INLA FITS

Hierarchical framework

3	**Hyperparameters**	$\theta \sim \pi(\theta)$, with θ = (range ρ, sd σ, noise precision τ)
2	**Latent Gaussian field**	$x \mid \theta \sim N(0, \Sigma(\theta))$, with $x = (\beta_0, \beta_1, \ldots, \beta_6, u)$
1	**Data**	$y \mid x, \theta \sim \pi(y \mid x, \theta)$

In our case: The log10-transformed data follow a Gaussian distribution.

The latent field follows a Gaussian distribution with Matérn covariance matrix:

$$\Sigma_{ij} = Cov(u(s_i), u(s_j)) = \sigma^2 / (2^{\nu-1}\,\Gamma(\nu)) \cdot (\kappa\|s_i - s_j\|)^{\nu} \cdot K_{\nu}(\kappa\|s_i - s_j\|)$$

Gaussian distribution also for the hyperparameters.

THE COMPARISON

MCMC works, but it pays for every step

MCMC: Gaussian random field (GRF)

Definition: $x(s)$ is defined at every location s. For any locations $s_1, \ldots, s_n$, the vector $x = (x(s_1), \ldots, x(s_n))^T \sim N(\mu, \Sigma)$, with

$$\pi(x) = \frac{1}{(2\pi)^{n/2}} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right), \quad x \in \mathbb{R}^n.$$

Σ fixes the correlation between every pair of observations.

Σ is $n \times n$ and dense and n = number of observations.

One factorisation $O(n^3)$.

INLA: Gaussian Markov random field (GMRF)

Definition: $x = (x_1, \ldots, x_m)^T \sim N(\mu, Q^{-1})$, on a graph $G = (V, E)$ with

$$\pi(x) = \frac{1}{(2\pi)^{n/2}} |Q|^{1/2} \exp\left(-\frac{1}{2}(x-\mu)^T Q (x-\mu)\right)$$

and

$$x_i \perp x_j \mid x_{-ij} \iff Q_{i,j} = 0 \iff \{i,j\} \notin E \quad \text{for all } i \neq j \text{ and } i,j \in \mathcal{V},$$

Markov: $x_i \perp x_j \mid x_{-ij}$ means

$$\pi(x_i, x_j \mid x_{-ij}) = \pi(x_i \mid x_{-ij})\, \pi(x_j \mid x_{-ij}).$$

Here x_{-ij} is x without x_i and x_j.

Q is $m \times m$ and sparse and m = number of nodes in G.

One factorisation $\approx O(m^{3/2})$.

THE SPDE TRICK

A Matérn field with a sparse precision matrix

SOME MATH: $(\kappa^2 - \Delta)^{\alpha/2}\, u(s) = \mathcal{W}(s), \qquad \mathcal{W} =$ white noise

Its solution is u(s) is a GRF with Matérn covariance function

$$\Sigma_{ij} \;=\; Cov(u(s_i), u(s_j)) \;=\; \sigma^2 \,/\, (2^{\nu-1}\,\Gamma(\nu)) \;\cdot\; (\kappa \| s_i \,-\, s_j \|)^{\nu} \;\cdot\; K_{\nu}(\kappa \| s_i \,-\, s_j \|)$$

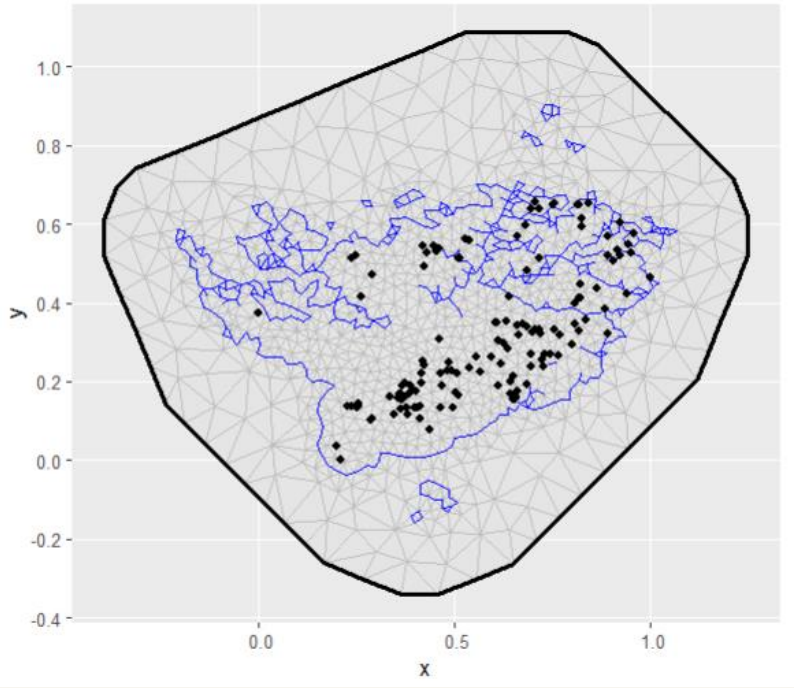
and smoothness: $\nu = \alpha - d/2$.
In 2D with $\alpha = 2$: $\nu = 1$ and range $\rho = \sqrt{8}/\kappa$.

Solve it with finite elements on a triangle mesh:

$$u(s_i) \;=\; \Sigma_k\, \psi_k(s_i)\, w_k \;=\; \Sigma_k\, A_{ik}\, w_k$$

or for all stations at once: $\boldsymbol{u} \;=\; \boldsymbol{A}\, \boldsymbol{w}, w \sim N(0, Q^{-1}),\; \Sigma = \; Q^{-1}$ and w a GMRF.

Each w_k interacts only with nearby nodes, so Q is sparse.
ψ_k are hat functions on the triangles.



THE WORKFLOW

Eight steps from points to a validated map

(1) **Mesh** — fm_mesh_2d()

(2) **A matrix** — fm_basis

(3) **SPDE, formula** — pcmatern, f()

(4) **Stack** — inla.stack

(5) **Fit** — inla()

(6) **Model output**

(7) **Model plots** — with INLAutils

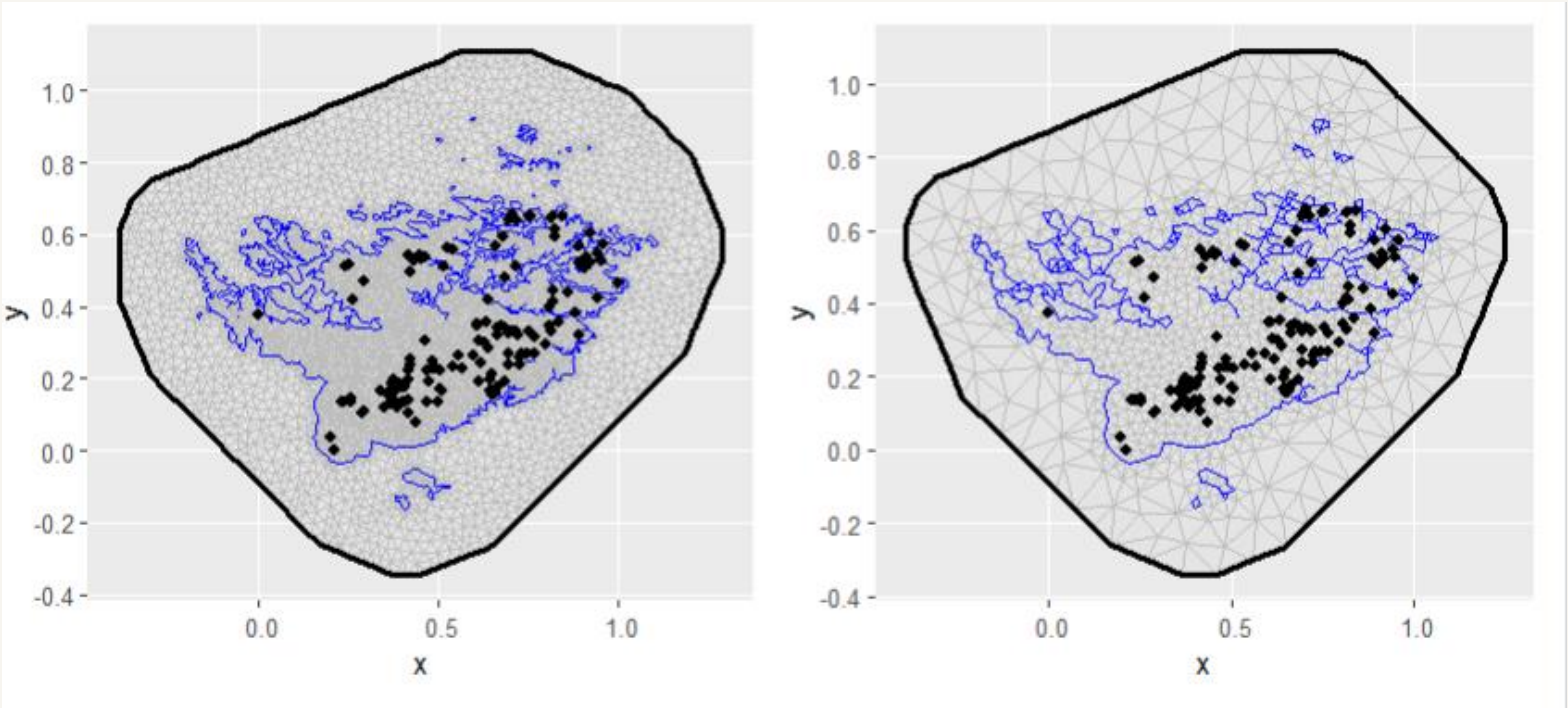
(8) **Cross-validate** — cpo, group.cv

Steps (1)-(4) only prepare inputs. (5) is the spatial inference call and (6)-(8) are post-processing steps.

Continue model_setup.R file to work through Step 1 to 8.

(1) MESH

The mesh decides where the field lives



Planar meshes need planar coordinates, not longitude and latitude.
Left mesh: 8825 vertices. Right mesh: 1264 vertices.

Fine where the data are, coarse around them

```r
build_mesh_ensemble <- function(coords, boundary, response,
                                n_levels = 6) {

  est       <- estimate_spatial_range(coords, response)
  emp_range <- mean(c(est$rho_exp, est$rho_mat))

  inner_edge_vals <- seq(emp_range / 3, emp_range, length.out = n_levels)

  cat(sprintf("Effective range: %.0f m\n", emp_range))
  cat("Inner edge levels:", round(inner_edge_vals), "m\n")

  #build the mesh using the boundary object dynamically
  mesh_list <- lapply(inner_edge_vals, function(inner) {
    fm_mesh_2d(
      #loc.domain = coords,    #your coords if you do not set boundary (but then only rough polygon)
      boundary   = boundary,  #sets the inner domain and outer ring is built from offset[2]
      #Inner edges run from range/6 to range/2. The usual rule is range/5 to range/10, so only your finest level meets it.
      max.edge   = c(inner/2, inner*1.5),      # longest triangle edge inside and in the outer ring
      cutoff     = inner / 5,  #merges input points closer than this, max.edge/5 is common
      offset     = c(inner * 1, emp_range * 2)  #Width of the inner and outer extension.
    )
  })

  cat("Built", length(mesh_list), "meshes | nodes:",
      sapply(mesh_list, function(m) m$n), "\n")

  list(mesh_list = mesh_list, inner_edge_vals = inner_edge_vals,
       emp_range = emp_range)
}
```

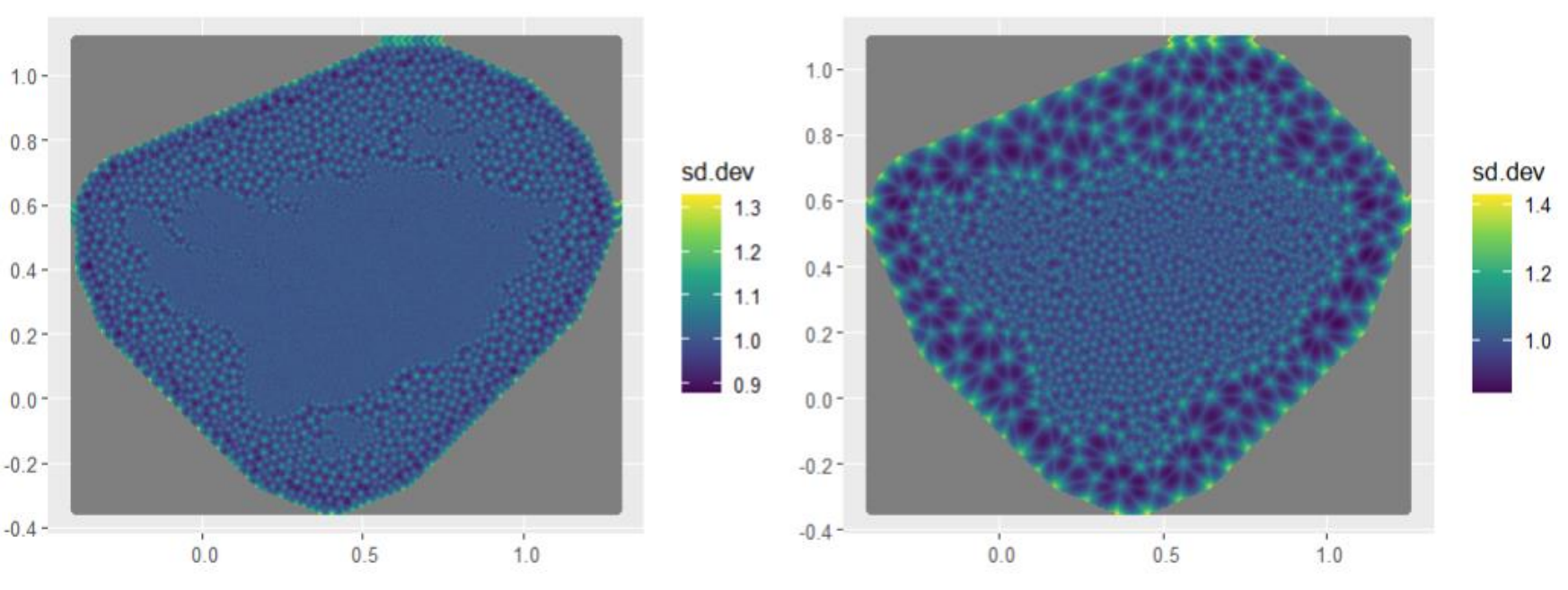
max.edge: inner edges of triangles ≈ range / 5 to range / 10

cutoff: Merges clustered points, no tiny triangles

offset: At least one range wide: boundary effects stay out

Range unknown?
estimate_spatial_range()

Ideal: Marginal SD is 1 everywhere inside the domain



Values of marginal SD of Matern field above 1 show boundary effects.
Test with inla.mesh.assessment()

(2) A MATRIX

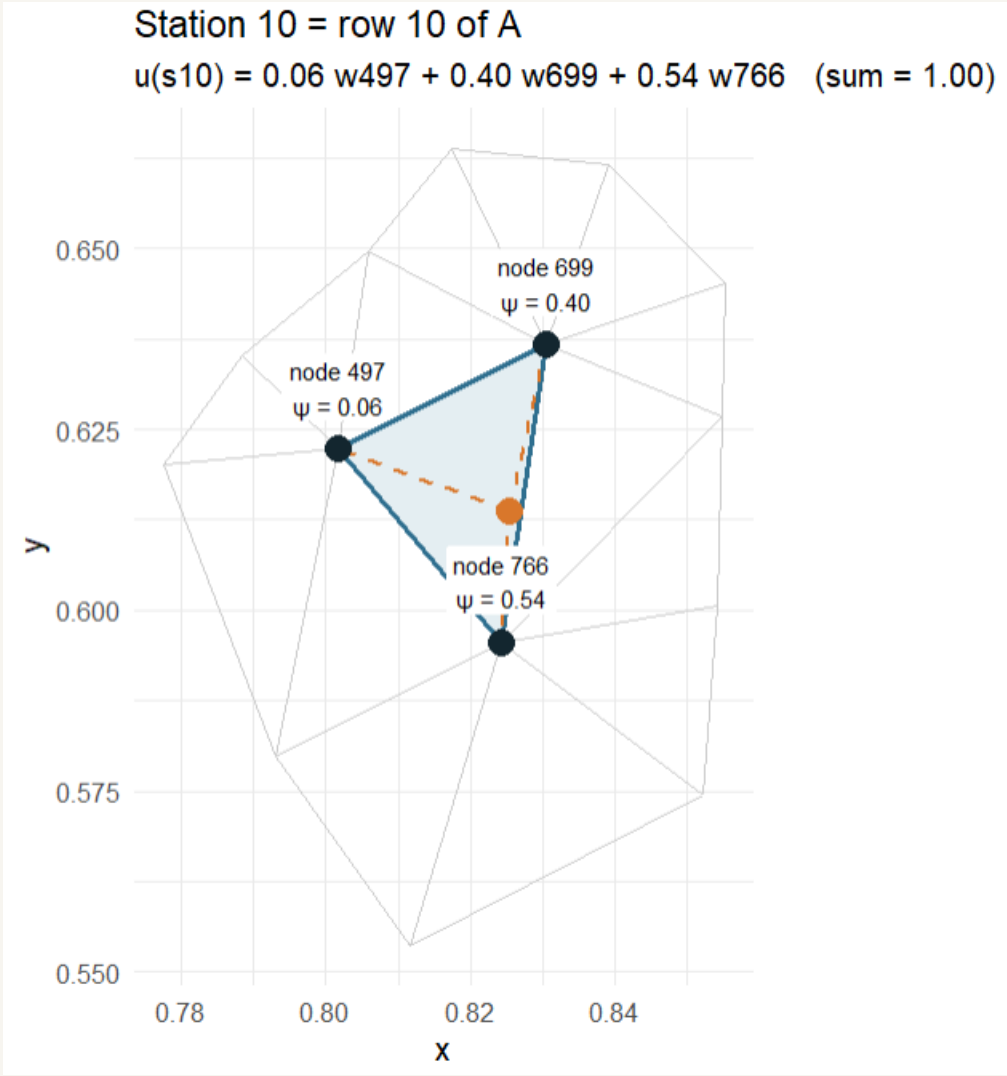
A links mesh vertices to locations

The field is stored only at the 1264 mesh nodes, as $w_1, .., w_{1264}$, with $w \sim N(0, Q^{-1})$.

Each w_k interacts only with nearby nodes, so Q is sparse.
ψ_k are hat functions: equal to 1 at node k, 0 at every other node and linear on each triangle.

Between the nodes the field is the surface
$u(s) = \Sigma_k \psi_k(s) w_k$.
A evaluates this sum at the stations: $A_{ik} = \psi_k(s_i)$.

A station inside a triangle has 3 non-zero values $\psi_k(s_i)$, one per corner.
They sum to 1, and the closer corner gets the larger value.



One row of A. At most 3 non-zeros, summing to 1.

PC priors speak in range ρ and SD σ

```
spde <- inla.spde2.pcmatern(
  mesh,
  prior.range = c(12000, 0.5), # P(range < 12 km) = 0.5
  prior.sigma =  c(0.3,0.5)    # P(sigma > 0.3) = 0.5, alternative: P(sigma > 1) = 0.01
  alpha = 2)
```

The SPDE needs (penalised-complexity (PC)) priors on range ρ and the SD σ => set them with two tail-probability statements:

$$P(\rho < \rho_0) = p_1 \; and \; P(\sigma > \sigma_0) = p_2,$$

$0 < p_1, p_2 < 1$ encode prior belief about the plausible correlation length and the field amplitude.

Our estimates come from estimate_spatial_range().
Alternatively: a field SD above 1 on the log10-scale is unlikely.

Why PC-priors? They shrink towards the simplest field: infinite range, zero SD.

(3) INDEX AND FORMULA

The formula names the field

```
index_space <- inla.spde.make.index(name = "spatial_field ", n.spde = spde$n.spde)

str(index_space ) # s = 1..1264, plus s.group and s.repl

formula <- y ~ -1 + Intercept + altitude + depth layer+ IDPR+ dist to river+ dist to fault+ dist to gypsum+ f(spatial_field, model = spde)
```

f(spatial_field, model = spde): A latent effect named spatial_field. Its values are the w_k at the mesh vertices.

-1+ Intercept: We pass our own intercept column. The stack needs every effect as a named variable.

s.group, s.repl: Unused today. They carry time or replicates in space-time models.

The stack lines up data, A matrices and effects

Linear predictor

$$\eta = A\,w + 1 \cdot (\beta_0 + \beta_1\,alt + \beta_2\,depth + \beta_3\,IDPR + \beta_4\,river + \beta_5\,fault + \beta_6\,gypsum)$$

tag	data	A (one per block)	effects (one per block)
est	y = log10(resultat), 146 rows	A (146 × 1264), 1	index_space, data.frame(Intercept, covariates)
pred	y = NA, 64500 rows	A_pred (64500 × 1264), 1	index_space, data.frame(Intercept, covariates)
joined	64646 rows	inla.stack.A(stk)	inla.stack.data(stk)

1 means identity matrix. A links the stations to the mesh nodes and 1 links them to their covariate row.
NA means predict. INLA returns the posterior for those rows.

(4) STACK

Two stacks, joined, found again by tag

```r
covar_names <- c("altitude", "order_min", "IDPR", "river_dist", "fault_dist", "dist_to_gypse")
```

```r
#observation stack
stk_est <- inla.stack(
  data = list(y = y_log),
  A = list(A_matrix, 1), # one projection matrix per effect block:
  # A_matrix for the field, 1 (identity) for the covariates
  effects = list(index_space,  # block 1: spatial field, indexed by mesh node
                 data.frame(Intercept = 1, stations[, covar_names])),
  tag = "est")  #through this tag you find the observation data in the stack

#for prediction: YOU NEED COVARIATE INFO AT EACH GRID POINT
stk_pred <- inla.stack(
  data = list(y = NA),   #prediction data NA (to be estimated)
  A = list(A_pred, 1),
  effects = list(index_space,
                 data.frame(Intercept = 1, pred_grid[, covar_names])),
  tag = "pred")  #through this tag you find the prediction data in the stack

#join stack
stk <- inla.stack(stk_est, stk_pred)
```

Typical errors:

- A and effects lists of different length;
- nrow(A) ≠ length(y_log);
- a formula variable missing from the effects.

(5) FIT

One call

```
res <- inla(
  formula,
  family = "gaussian",
  data = inla.stack.data(stk, spde = spde),
  control.predictor = list(
    A = inla.stack.A(stk),
    compute = TRUE, link = 1),
  control.compute = list(
    dic = TRUE, waic = TRUE, cpo = TRUE,
    config = TRUE,
    return.marginals.predictor = TRUE),
verbose=FALSE)

summary(fit) # about 6 s here
```

formula: the one we set before

family: the likelihood of the data, here "gaussian« ; many more are available, see inla.list.models("likelihood")

data: the variables from the stack

control.predictor: settings for linear predictor

- **A**: the stacked projection matrix, linking the mesh nodes to the stations and grid points
- **compute = TRUE**: returns posterior of every stack row
- **link = 1**: uses family link for the NA rows => predictions are on y scale

control.compute: additional calculations

- **dic, waic**: information criteria for comparing models
  cpo: leave-one-out predictive values (CPO and PIT) for cross-validation
- **config**: needed for posterior sampling with inla.posterior.sample() and for inla.group.cv()
- **return.marginals.predictor**: keeps the full posterior distribution for every row but uses a lot of memory with many prediction rows.

(5) FIT · MODEL COMPARISON

Without distance covariates, the field has to carry the trend

Posterior mean range:

0.08

with distance covariates · 95%: 0.04 to 0.15

Posterior mean range

0.19

without distance covariates · 95%: 0.09 to 0.35

Models (scaled)	DIC	WAIC
Intercept + all covariates + field	**-40**	**-35**
Intercept + altitude+ IDPR+ field	-35	-29

DIC (Deviance Information Criterion): how well does the model fit the data+ penalty for model complexity.

WAIC (Watanabe–Akaike Information Criterion): how well the model is expected to predict new data.
Average over the full posterior. Approximates leave-one-out cross-validation.

(6) READING THE OUTPUT OF BEST MODEL

Groundwater sulfate drops away from with altitude and distance to gypsum

Parameter	Mean	95% interval
Intercept β_0	1.57	1.42 to 1.73
altitude β_1	−0.14	−0.22 to −0.05
distance to gypsum β_6	−0.21	−0.3 to −0.12
Range rho ρ	0.08	0.04 to 0.16
Field SD σ	0.28	0.21 to 0.37
Noise SD	0.17	0.13 to 0.21

```
summary(res_scaled)

# noise sd from the precision
noise_sd <- inla.tmarginal( function(x) 1 / sqrt(x),
  res_scaled$marginals.hyperpar[[1]])

inla.zmarginal(noise_sd)
```

res_scaled$summary.linear.predictor:
posterior of $\eta_i = X\beta + A\,w$ for each row, on the scale of the link function $\mu_i = g^{-1}(\eta_i)$.
With a Gaussian likelihood g=identity function

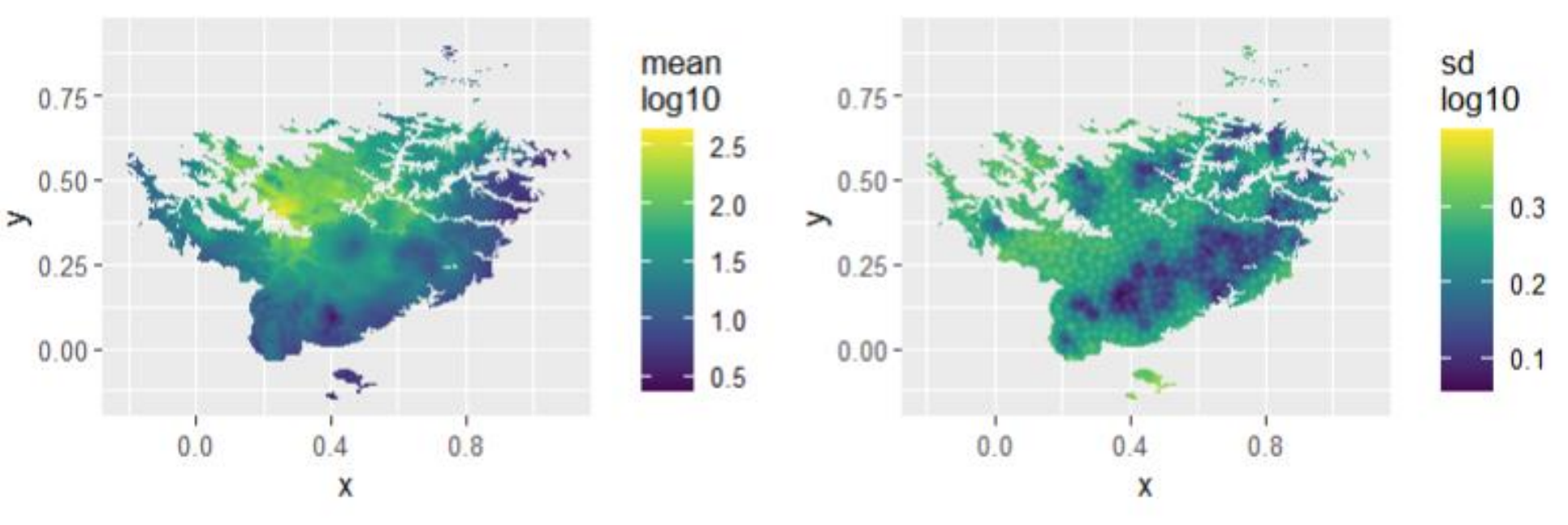
res _scaled$summary.fitted.values: posterior of the mean response at each stack row, on the data scale.

summary.hyperpar: posteriors of the hyperparameters, here the noise precision, the range and the field SD.

The prediction map and its uncertainty

```
pred_grid$pred_mean <- res_scaled$summary.fitted.values$mean[idx_pred]
pred_grid$pred_sd   <- res_scaled$summary.fitted.values$sd[idx_pred]

ggplot(pred_grid, aes(x, y, fill = pred_mean)) +  geom_raster()
```

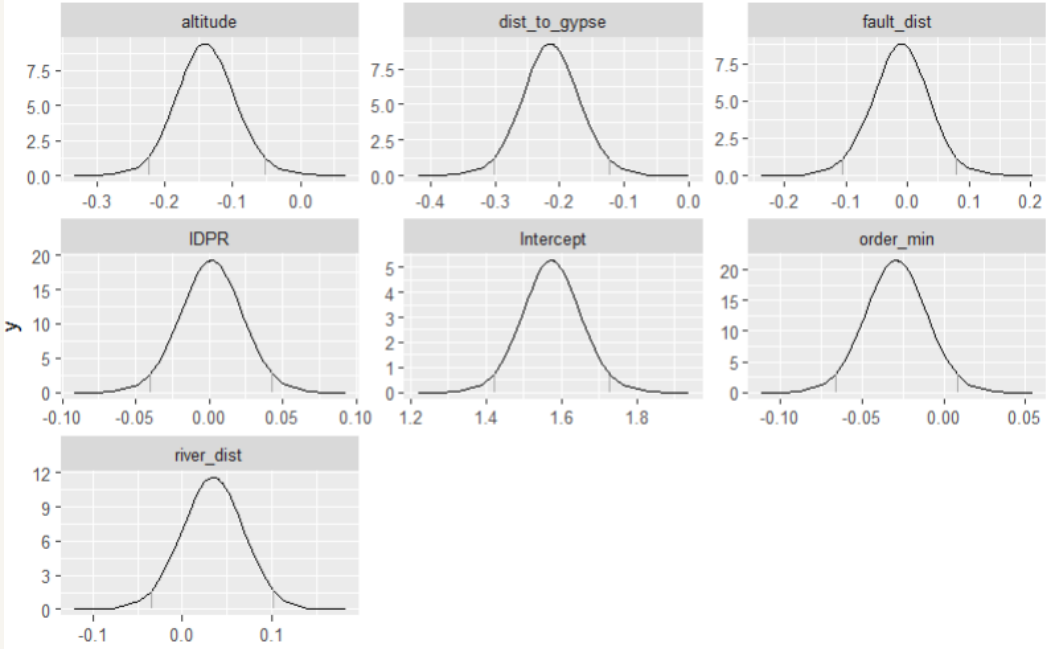


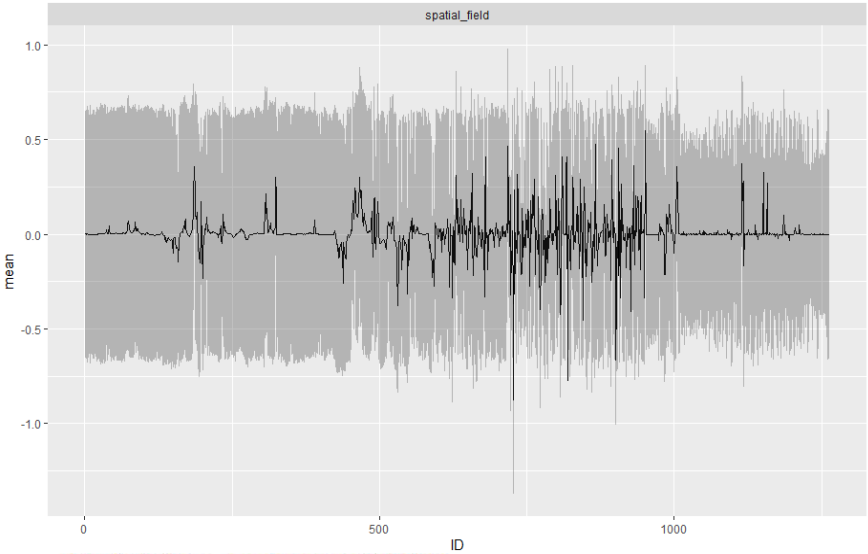
(7) MODEL PLOTS

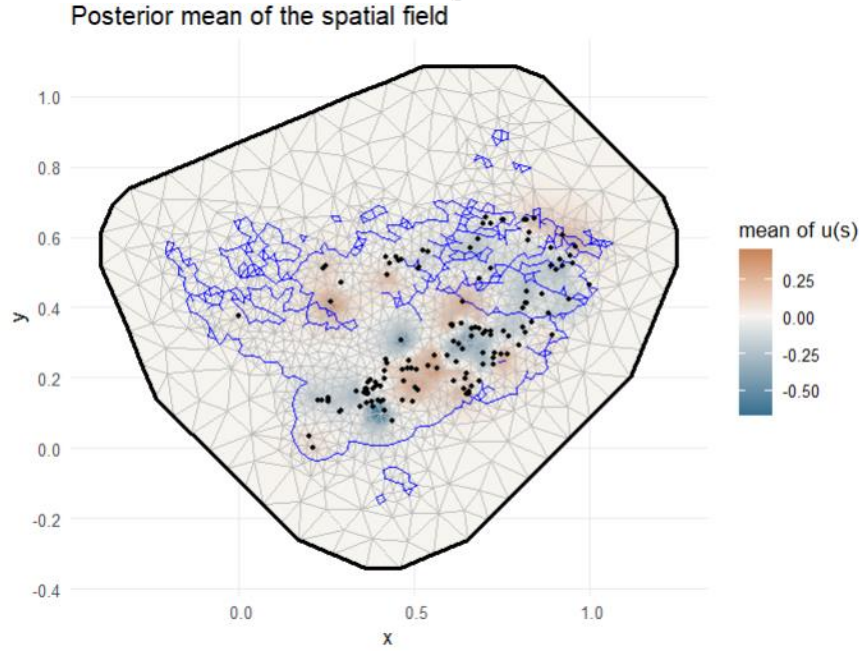
INLAutils diagnostic plots

```
autoplot(res_scaled, which = c(1, 2, 3))

ggplot_inla_residuals(res_scaled,
  observed = y_log, CI = TRUE)
```







(8) CROSS-VALIDATION

Two cross-validation schemes

1 · Leave-One-Out from the fit

CPO and PIT, computed during inla()

```
cpo <- res_scaled$cpo$cpo[idx_est]

-mean(log(cpo)) # log score

OR

loo<- inla.group.cv(res_scaled, num.level.sets= -1)
```

2 · Leave-group-out

Drops each point with its correlated neighbours

```
lgocv <- inla.group.cv( res_scaled, num.level.sets = 3)
-mean(log(lgocv$cv[idx_est]))
lgocv$groups[[1]]$idx

# groups chosen by distance (disc of radius rad)
D <- as.matrix(dist(coords.stations))

disc_groups <- function(rad)
lapply(seq_along(idx_est), function(i)
idx_est[D[i, ] <= rad])

rads <- c(0, 0.25, 0.5, 1, 1.5, 2) * my_range

disc_cv <- lapply(rads, function(r)
inla.group.cv(res_scaled, groups = disc_groups(r)))
 sapply(disc_cv, function(g) -mean(log(g$cv[idx_est])))
```

Scores: log score from CPO, then RMSE, CRPS and 95% coverage for the folds. **Check cpo$failure before you trust CPO.**

CHECKLIST

Eight checks before you trust a map

- Projected coordinates in sensible units, such as km
- Outer extension of mesh at least one range wide
- Rows found by tag with inla.stack.index(), never by position
- Priors set on interpretable scales, and a sensitivity refit
- Every prediction location inside the mesh: fm_is_within()
- spde passed to inla.stack.data()
- cpo$failure checked before using CPO
- Spatial CV when the map is used away from the data

WHERE NEXT

The same eight steps carry much further

Other likelihoods

Counts, presence or absence, positive data: change family to "poisson", "binomial" or "gamma".
Do not forget about the link function!

Space and time

The group argument links one field per time step, for example through an AR(1).
Either by adding f(time_index, model="rw1") or by grouping f(spatial_field, model=spde, group=year, control.group=list(model="ar1"))

inlabru

Builds the A matrices and stacks for you, and allows non-linear predictors.

Barriers

Coastlines and other physical barriers that correlation should not cross.

CONTACT INFO

Dr. Corinna Perchtold

- LinkedIn: https://www.linkedin.com/in/corinna-perchtold/
- Website: https://sites.google.com/view/cperchtold
- Reach out to me if you have questions or want to collaborate on an INLA project!

REFERENCES

Further reading

Cressie (1994). Statistics for spatial data. Wiley.

Lee & Nelder (1996). Hierarchical generalized linear models. JRSS B 58(4), 619–678.

Rue, Martino & Chopin (2009). Approximate Bayesian inference for latent Gaussian models using integrated nested Laplace approximations. JRSS B 71(2), 319–392.

Lindgren, Rue & Lindström (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: the SPDE approach. *JRSS B* 73(4), 423–498.

Krainski et al. (2018). *Advanced Spatial Modeling with Stochastic Partial Differential Equations Using R and INLA.* CRC Press.

Righetto, Faes, Vandendijck & Ribeiro (2020). On the choice of the mesh for the analysis of geostatistical data using R-INLA. *Commun. Stat. Theory Methods* 49(1), 203–220.

Fuglstad, Simpson, Lindgren & Rue (2019). Constructing priors that penalize the complexity of Gaussian random fields. *JASA* 114(525), 445–452.

Lucas, Python & Redding (2020). Graphical outputs and spatial cross-validation for the R-INLA package using INLAutils. arXiv:2004.02324.

Liu, Van Niekerk & Rue (2025). Leave-group-out cross-validation for latent Gaussian models. *SORT* 49(1), 121–146.

Adin, Krainski, Lenzi, Liu, Martínez-Minaya & Rue (2024). Automatic cross-validation in structured models: is it time to leave out leave-one-out? Spatial Statistics 62, 100843.